

Phase 2 Production Bug Fix Report

Sales Mode Activation — Vyapar AI Employee Platform

Status: FIXED PRODUCTION BUG

Date: June 2, 2026

Environment: Telegram + Render (company_id=hons , user_id=2071979418)

1. Problem Summary

Phase 2 lead detection and persistence worked correctly in production, but **sales mode never activated**. The bot replied like a FAQ/general assistant instead of a sales employee.

Component	Production behavior	Expected
Lead detection	Working — LEAD_SIGNALS_DETECTED	Working
Lead upsert	Working — score 20→50, stage interested	Working
Memory	Working — name recall confirmed	Working
Sales mode	Always sales_mode=False	sales_mode=True when qualifying
SALES_MODE_ACTIVE log	No matching logs	Log on every sales reply

User conversation examples (all failed sales mode)

- Kolhabi ma HONS ko net chha?
- Banepa ma internet jodnu chha
- Malai 500 Mbps ko net chahiyo
- Mero phone number dinchhu
- Chhito install garna parcha

2. Root Cause

Primary: `_resolve_sales_mode()` in `ai_employee_engine.py` was too narrow. It ignored lead stage, lead score, and active lead context.

Production logs showed the mismatch clearly:

Log field	Production value	Why sales_mode stayed False
buying	False	Nepali chha not matched (patterns used <code>cha</code> only)
stage	interested	Not used in sales_mode logic
score	40–50	Not used in sales_mode logic
detected_intent	often <code>general</code>	Blocked the intent-based path
SALES_MODE_ACTIVE	never logged	Only logged inside <code>sales_mode</code> and <code>exact_product</code> is <code>None</code>

Old logic only returned `True` when:

- `bundle.buying_intent == True` , OR
- `detected_intent` ∈ {`buying_intent`, `sales`, `pricing`, `coverage_inquiry`} AND (speed mismatch OR coverage)

So `should_process_lead()` saved leads, but prompts still used full catalog FAQ mode.

Secondary: Nepali pattern gaps

User message	Pattern issue
internet jodnu chha	Pattern expected cha , not chha
install garna parcha	Not in buying intent patterns
number dinchhu	Pattern was dinchu only
Kolhabi ma HONS ko net chha?	Not detected as coverage inquiry

3. Fix Applied

3.1 ai_employee_engine.py — sales mode logic

`_resolve_sales_mode()` now also activates when:

- `lead.stage` is `interested`, `qualified`, or `hot`
- `lead_score` ≥ 40 (max of persisted lead + current bundle)
- `coverage_check_needed`
- `should_process_lead` + unavailable speed or urgency signal

Returns `(bool, reason)` and **always** logs `SALES_MODE_ACTIVE` when true:

```
SALES_MODE_ACTIVE user_id=... company_id=hons reason=lead_stage=interested stage=interested score=50
```

3.2 lead_extractor.py — pattern fixes

- Buying intent: `chh?[au]` variants (`chha/chu/cha`)
- Buying intent: `install garna parcha`
- Contact: `number dinchhu`
- Coverage: `ko net chha`, `net chha?`, location + `?` questions

3.3 intent_engine.py — aligned patterns

Buying and coverage intent detection updated to match Nepali Telegram usage.

3.4 tests/test_sales_mode.py — new tests

- `chha` variant buying intent
- `install parcha` buying intent
- Interested lead stage activates sales mode
- Score ≥ 40 activates sales mode
- Coverage question activates sales mode

4. Files Changed

File	Change
<code>ai_employee_engine.py</code>	Expand <code>_resolve_sales_mode</code> , add reason logging
<code>lead_extractor.py</code>	Nepali <code>chha/coverage/contact</code> patterns
<code>intent_engine.py</code>	Aligned buying/coverage patterns
<code>tests/test_sales_mode.py</code>	New unit tests

5. Testing Checklist

#	Telegram message	Expected after fix
1	Banepa ma internet jodnu chha	<code>buying=True</code> , <code>SALES_MODE_ACTIVE</code> <code>reason=buying_intent</code>
2	Malai 500 Mbps ko net chahiyo	<code>sales_mode=True</code> , single alternative (no full catalog)
3	Kolhabi ma HONS ko net chha?	Coverage-aware sales reply, asks for contact
4	Mero phone number dinchhu	Contact signal; sales mode if lead already interested
5	Chhito install garna parcha	<code>buying=True</code> , urgency high, sales mode
6	Support hours kati ho?	No lead, no sales mode
7	Mero naam ke ho?	Direct memory answer, no sales mode

Run locally before deploy:

```
python -m unittest tests.test_sales_mode tests.test_lead_phase2 -v
```

6. Expected Logs After Fix

```
LEAD_SIGNALS_DETECTED user_id=2071979418 company_id=hons buying=True score=50 stage=interested signals={...}
LEAD_UPSERTED user_id=2071979418 company_id=hons lead_id=1 stage=interested score=50
SALES_MODE_ACTIVE user_id=2071979418 company_id=hons reason=lead_stage=interested stage=interested score=50
EMPLOYEE_REPLY_GENERATED user_id=2071979418 company_id=hons sales_mode=True
```

Possible SALES_MODE_ACTIVE reason values

reason	Trigger
buying_intent	Explicit buying language detected
lead_stage=interested qualified hot	Active lead in qualification stage
lead_score=40 (or higher)	Score threshold reached
coverage_check_needed	Area availability check required
no_exact_product_match	Requested speed not in catalog
unavailable_speed	Speed signal + no exact match
urgency_signal	High/medium urgency detected

7. Deploy

Deploy these files together on Render (atomic deploy):

- ai_employee_engine.py
- lead_extractor.py
- intent_engine.py

No new environment variables required. Restart service and re-run Telegram smoke tests above.